CUSTOMERS LIKE YOU

Mini-Project #01 | PCA vs. Truncated SVD for Retail Recommendations

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Subject: Recommendation model and customer segmentation findings		

Executive recommendation

Advance **Truncated SVD** as the primary candidate for further validation, while retaining PCA as a comparison. The purchase matrix is highly sparse, and Truncated SVD works directly with sparse data without centering it. In this run, SVD fitted in 0.163 seconds versus 0.838 seconds for PCA (about 5.1 times faster). Its Hit Rate@10 was 20.02%, close to PCA at 20.13%, and both exceeded the best-seller baseline of 19.29%. These results support continued testing, not immediate production deployment.

What we built and what compression changes

The pipeline loaded 1,067,371 invoice lines from both Online Retail II sheets. After removing cancellation invoices, non-positive quantity or price, non-product codes, lines without customer IDs, and duplicate rows, 776,577 rows remained. A time-based split produced 660,194 training rows and 71,393 test rows, with a cutoff of October 9, 2011 at 12:50. The training purchase matrix contains 5,473 customers and 3,687 products, with 413,927 non-zero entries and 97.95% sparsity.

The matrix would require about 161.43 MB as a dense array, while its sparse CSR representation uses about 3.333 MB. At k = 50, the stored customer and product factors occupy about 1.832 MB. Compression replaces the original high-dimensional purchase patterns with a smaller set of latent factors. This makes similarity-based recommendations more practical, but it also loses some detail from the original purchase matrix.

Evaluation and customer groups

Hit Rate@10 measures the share of eligible test customers for whom at least one held-out purchase appears in the model's top 10 recommendations. Across 1,913 eligible test customers, Truncated SVD achieved 20.02%, PCA achieved 20.13%, and the best-seller baseline achieved 19.29%. The gap between PCA and SVD is small; SVD's main advantage in this run is fit speed and its natural fit for sparse data. The customer map produced four groups: Retail Mix Segment 1 (1,679 customers), Candle & Homeware Buyers (835), Retail Mix Segment 3 (2,173), and Bag & Packaging Buyers (786). Leading products included cake cases and gliders, heart-shaped candle holders and bird ornaments, ceramic storage jars and mugs, and jumbo bags and lunch bags, respectively.

Risks, limitations, and next steps

- **Modest improvement:** SVD beats the baseline by 0.73 percentage points in Hit Rate@10; this is encouraging but not a large gain.
- **Information loss:** SVD's estimated relative reconstruction error is 0.58405 at k = 50. PCA explains 65.056% of variance at the same k; these are different metrics and should not be treated as directly equivalent.
- **Measurement limits:** Python tracemalloc may not capture all native-library memory; the reported reconstruction error is an estimate.
- **Validation:** test additional chronological holdouts, other component counts, raw versus binary versus log-transformed purchase values, segment stability, and cold-start cases. Review the segments with Marketing before deployment.

Decision: Continue with Truncated SVD as the primary candidate for validation. Do not deploy yet; confirm that recommendation quality and customer segments remain useful across additional time periods and data representations.